



Project Presentation of Industry Oriented Hands on Experience (22CS422)

On

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Research Overview

Semester - 8

Title: Comparative Analysis of Spatial, Edge-Preserving and Transform-Based Filters for Medical Image Enhancement

Focus: A comprehensive evaluation of spatial, edge-preserving, and transform-based filters for enhancing CT, MRI, and ultrasound image quality

Why Medical Image Denoising Matters



The Challenge

Noisy images pose significant challenges for radiologists, doctors, and medical specialists. Different denoising techniques exhibit varying performance depending on noise type, imaging modality, and computational constraints.

- Medical imaging modalities like MRI, CT, and Ultrasound are essential for disease diagnosis, surgical planning, and treatment monitoring
- These images are often degraded by noise introduced during acquisition, transmission, or reconstruction processes
- Noise represents unwanted random signals that reduce image quality and obscure important diagnostic information
- Noise can affect the accurate identification of diseases such as tumors
- Effective denoising is essential to enhance image quality
- Denoising helps in preserving critical anatomical structures and important diagnostic details

Three Categories of Denoising Techniques



Spatial Domain Filters

Mean, Median, Gaussian, Moving Average filters that operate directly on pixel values

Edge-Preserving Filters

Bilateral, Anisotropic Diffusion, Non-Local Means that maintain structural boundaries

Transform-Based Methods

Wavelet, Curvelet, Wiener filters that operate in frequency domain

Each category offers distinct advantages and trade-offs between noise suppression, structural preservation, and computational efficiency.

Dataset and Methodology

Imaging Modalities

Experiments conducted on three publicly available datasets:

- **MRI Dataset:** Brain Tumor MRI Dataset from Kaggle with tumor types and non-tumor cases
- **CT Dataset:** COVID-19 CT Scan Lesion Segmentation Dataset with lesion annotations
- **Ultrasound Dataset:** Breast Ultrasound Images Dataset categorized into benign, malignant, and normal classes

Evaluation Metrics

- Mean SquaredError (MSE)
- Peak Signal-to-Noise Ratio (PSNR)
- Structural Similarity Index (SSIM)
- Signal-to-Noise Ratio (SNR)



Understanding Medical Image Noise



1

Gaussian Noise

Statistical noise from electronic fluctuations during image acquisition. Affects almost every pixel with slight intensity variations. Common in MRI and CT images.

2

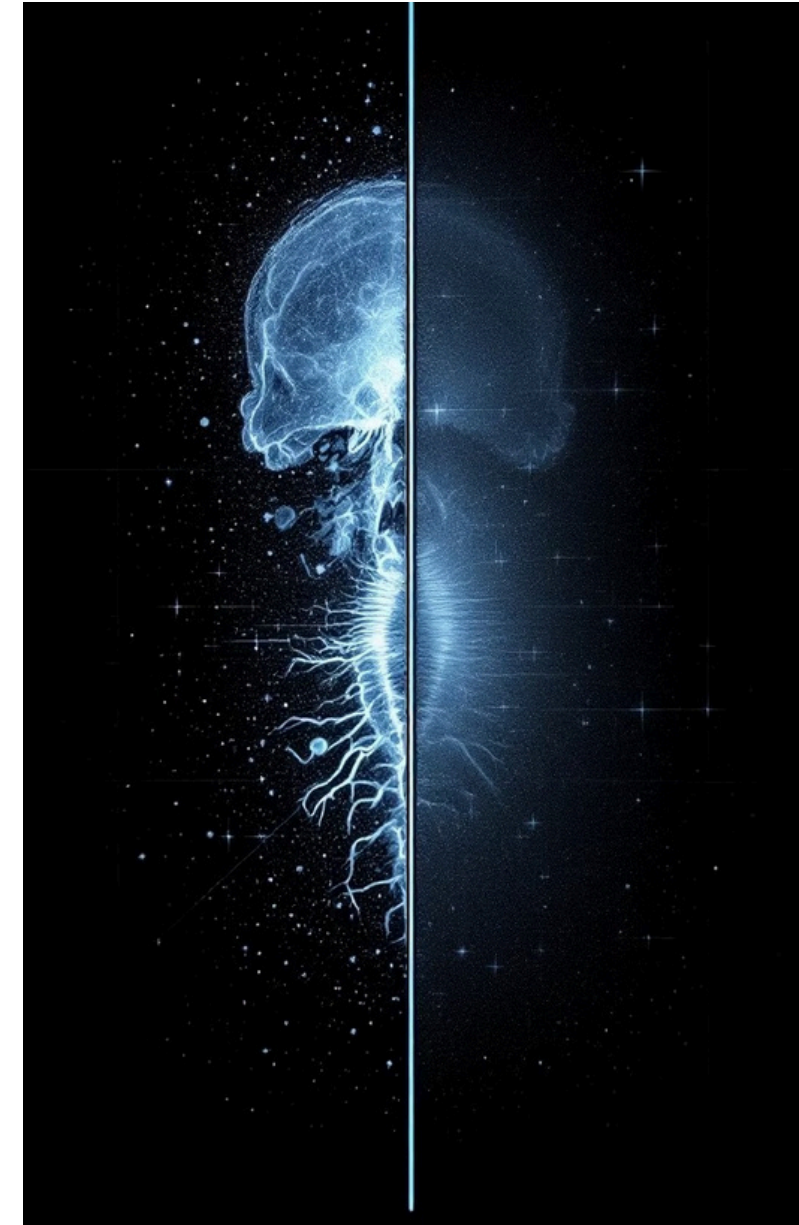
Salt-and-Pepper Noise

Impulse noise appearing as randomly distributed bright and dark pixels. Caused by transmission errors or sensor malfunction. Significantly distorts image details.

3

Speckle Noise

Multiplicative noise from interference of reflected signals in ultrasound imaging. Produces granular appearance that reduces contrast and obscures boundaries.



Key Findings from Analysis



Spatial Filters

Computationally efficient but may blur edges and reduce fine structural details



Edge-Preserving

Better structural preservation with Bilateral filter consistently performing best

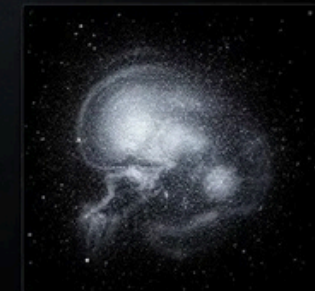
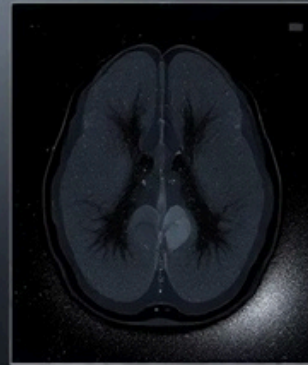
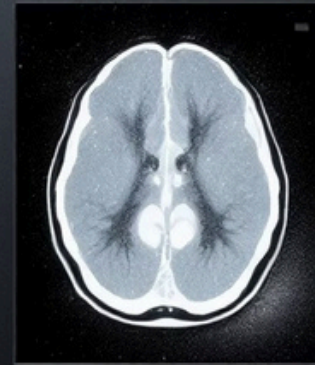


Transform Methods

Wiener filter emerged as most effective across all modalities



Critical Insight: No single denoising technique performs optimally for all imaging modalities. Selection should be guided by noise characteristics and structural requirements of specific medical imaging applications.



BEFORE

CLEANER



Tumor Segmentation

Enhanced MRI image quality improves accuracy of tumor detection and boundary identification in brain scans



Fetal Imaging

Ultrasound denoising provides clearer visualization for prenatal diagnosis and fetal development monitoring



3D Reconstruction

Improved image quality enables better 3D modeling for surgical planning and implant generation



Fracture Detection

Improved CT image clarity enables better identification of skull fractures and bone abnormalities



Retinal Analysis

Enhanced vessel visualization supports early detection of diabetic retinopathy and other eye diseases



AI Diagnostics

Cleaner images enhance performance of automated machine learning pipelines for computer-aided diagnosis



Key Conclusions

- Denoising performance varies significantly across different imaging modalities and noise types, making it important to choose techniques based on the specific use case
- Spatial filters offer computational efficiency, edge-preserving methods help maintain structural details, and transform-based approaches provide a balanced trade-off between noise reduction and detail preservation
- The study enhances understanding of denoising techniques and supports the development of more reliable medical image processing systems for accurate diagnostic and clinical applications

Future Research

- Adaptive parameter tuning mechanisms
- Comparison with deep learning methods (DnCNN, U-Net)
- 3D volumetric denoising for CT/MRI volumes
- Real-time implementation for clinical workflows



*Thank
You*